GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

# Model Paper — 03

Third Term Test (Full Syllabus)

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**Time:** 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

**Instructions:**

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of $\pi$ as $\frac{22}{7}$. The use of calculators is not allowed.

| Part | A | B1 | B2 | B3 | B4 | B5 | B6 | Total |
|---|---|---|---|---|---|---|---|---|
| Marks | /40 | /10 | /10 | /10 | /10 | /10 | /10 | /100 |

## Part A

*Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)*

**1.** Find the square root of 7744 using the division method.

Answer: .....

**2.** Simplify: $(1\frac{3}{4} + 2\frac{1}{2}) \div 1\frac{7}{10}$

Answer: .....

**3.** Expand and simplify: $(x + 3)^2 - (x - 3)^2$

Answer: .....

**4.** Factorise: $3x^2 + 10x - 8$

Answer: .....

**5.** Find the least common multiple of the expressions $x^2 - 4$ and $x^2 + x - 6$.

Answer: .....

**6.** Simplify: $\frac{2x}{x^2 - 1} \div \frac{4x^2}{x + 1}$

Answer: .....

**7.** Find the length of the arc of a sector of radius 10.5 cm and central angle 60°.

Answer: .....

**8.** The angles of a triangle are in the ratio 2 : 3 : 4. Find the size of the largest angle.

Answer: .....

**9.** If $\log_3 (x + 1) = 2$, find the value of x.

Answer: .....

**10.** Write as a single logarithm and simplify: $2 \lg 3 + \lg 5 - \lg 9$

Answer: .....

**11.** Find the sum of the first 20 terms of the arithmetic progression 2, 5, 8, ...

Answer: .....

**12.** If $n(\xi) = 50$, $n(A) = 30$, $n(B) = 25$ and $n(A \cap B) = 12$, find $n(A \cup B)'$.

Answer: .....

**13.** The price of an article was increased by 20%, and then the new price was reduced by 20%. Find the percentage change relative to the original price.

Answer: .....

**14.** A car travelling at 60 km per hour takes 4 hours for a journey. Find the speed required to complete the same journey in 3 hours.

Answer: .....

**15.** Make $l$ the subject of the formula $T = 2\pi\sqrt{\frac{l}{g}}$.

Answer: .....

**16.** Solve the inequality $5 - 2x \leq 11$ and write the least integer value that x can take.

Answer: .....

**17.** One card is drawn at random from 20 identical cards numbered 1 to 20. Find the probability that its number is a multiple of 4.

Answer: .....

**18.** The mean of 5 numbers is 12. Find the mean of the remaining numbers when the number 20 is removed from them.

Answer: .....

**19.** In the quadrilateral ABCD, AD ∥ BC. The area of triangle ABC is 35 $cm^2$. If E is the midpoint of BC, find the area of triangle DEC.

Answer: .....

**20.** A tap that supplies 12 litres per minute is used to fill an empty tank of capacity 0.9 $m^3$. Find the time taken to fill the tank. (1 $m^3$ = 1000 litres)

Answer: .....

## Part B

*Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)*

**1.** (a) Simplify: $\frac{x^2 - x - 6}{x^2 - 9} \times \frac{x^2 + 3x}{x^2 + 2x}$ (3 marks)

(b) The sum of two numbers is 47. Twice the larger number exceeds three times the smaller number by 9. Taking the larger number as x and the smaller number as y, form a pair of simultaneous equations, solve them and find the two numbers. (4 marks)

(c) Solve: $3x^2 - 7x + 2 = 0$ (3 marks)

**2.** An incomplete table of values prepared for drawing the graph of the function $y = 2x^2 - 3$ is shown below.

| **x** | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| **y** | 15 | ..... | −1 | −3 | ..... | 5 | 15 |

(a) Complete the blanks in the table. (1 mark)

(b) Draw the graph taking 10 small squares to represent 1 unit along the x-axis and 10 small squares to represent 2 units along the y-axis. (3 marks)

(c) Using the graph,

(i) write the coordinates of the turning point of the function. (1 mark)

(ii) write the approximate roots of the equation $2x^2 - 3 = 0$. (2 marks)

(iii) write the range of values of x for which $y < 5$. (1 mark)

(iv) write the range of values of x for which y decreases as the value of x increases. (1 mark)

(d) Write the function of the graph obtained when the graph is moved along the y-axis so that it passes through the origin. (1 mark)

**3.** The cross-section of a solid metal prism is a right-angled triangle whose sides containing the right angle are 7 cm and 24 cm. The length of the prism is 11 cm.

(a) Find the volume of the prism. (2 marks)

(b) Find the length of the hypotenuse of the triangular cross-section and hence find the total surface area of the prism. (3 marks)

(c) This prism is melted and a solid cylinder of radius 7 cm is made. If no metal is wasted, find the height of the cylinder. (3 marks)

(d) Find the area of the curved surface of that cylinder. (2 marks)

**4.** A point E lies on the side DC of a parallelogram ABCD. AE and BE are joined.

(a) Draw a rough sketch including the above information. (1 mark)

(b) Show, with reasons, that the area of triangle ABE = ½ × the area of parallelogram ABCD. (2 marks)

(c) Show that the sum of the areas of triangles ADE and BCE is equal to the area of triangle ABE. (2 marks)

(d) If the area of ABCD is 84 $cm^2$ and DE : EC = 3 : 4, find the areas of triangles ADE and BCE separately. (3 marks)

(e) If G is the midpoint of AE, find the area of triangle DGE. (2 marks)

**5.** The expenses of a family with a monthly income of Rs. 60 000 are shown below. The remaining money is deposited as savings.

| Item | Food | Education | Transport | Electricity and water | Savings |
|---|---|---|---|---|---|
| Amount (Rs.) | 24 000 | 12 000 | 9 000 | 6 000 | ..... |

(a) Find the amount for savings and calculate the angle at the centre corresponding to each item for drawing a pie chart. (3 marks)

(b) Represent this information in a pie chart within a circle of radius 4 cm. (3 marks)

(c) Write the amount spent on food as a percentage of the total income. (1 mark)

(d) In a later month, the food expense rose by 15%, and that increase was deducted from the savings. (The income and the other expenses remain unchanged.) Find the angle at the centre for savings in the new pie chart. (3 marks)

**6.** A contractor agreed to build a wall in 30 days using 20 workers. After working for 12 days, only ⅓ of the wall had been completed. (Assume that all the workers work at the same rate.)

(a) Find the number of man-days required to build the whole wall. (3 marks)

(b) If all 20 workers continue working, how many more days will it take to complete the remaining work? By how many days will the completion of the wall be delayed? (2 marks)

(c) Find the minimum number of workers that must be newly employed after the 12 days in order to complete the wall within the agreed 30 days. (3 marks)

(d) If only a total of 25 workers can be obtained after the 12 days, by how many days will the completion of the wall be delayed? (2 marks)

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